

Probability and Conditional Probability

Example 1

You have two books. You have a probability of 0.3 of liking book A, and a probability of 0.4 of liking book B. You have a probability of 0.2 of liking both book A and B. What is the probability that you like neither book?

Answers

Let A be the event that you like book A, and B be the event that you like book B.

$$P(A) = 0.3 \quad P(B) = 0.4 \quad P(AB) = 0.2$$

$$P(A \cup B) = P(A) + P(B) - P(AB) = 0.3 + 0.4 - 0.2 = 0.5$$

The event that you like neither book is $(A \cup B)^c = A^c B^c$.

$$P(A^c B^c) = 1 - 0.5 = 0.5$$

Example 2

The game of ‘Toto’ is a football match outcome guessing game played in some European countries. There are 13 football matches and hence 13 entries. In each entry, you can tick it is a ‘home win’, ‘draw’ or ‘away win’.

The prize money depends on how many of your guesses are correct.

If you are just guessing,

- a) what is the probability that all your 13 guesses are correct?
- b) what is the probability that 11 of your guesses are correct?

Answers

$$P(13 \text{ hits}) = \left(\frac{1}{3}\right)^{13} = 6.272254772 \times 10^{-7} \approx 0.000000627$$

(about 0.5 millionth)

For 11 hits, the problem is equivalent to selecting 11 positions out of 13 positions. There are $\binom{13}{11}$ ways. Hence

$$P(11 \text{ hits}) = \binom{13}{11} \left(\frac{1}{3}\right)^{11} \left(\frac{2}{3}\right)^2 = 1.95694348 \times 10^{-4} \approx 0.0002$$

(about 1 in 5000)

Example 3

Probability is readily applied in games of chances.

Poker is played with a 52 card deck (13 cards of 4 suits). A player's hand is composed of 5 cards.

A 'royal flush' is the most difficult hand to obtain. It consists of Ace, King, Queen, Jack and 10, all of the same suit.

A 'pair' is the hand of the lowest value. It contains a simple pair of cards of the same value. An example is Ace Ace Four Jack Nine.

When the cards are dealt at the beginning of the game,

- a) what is the probability of obtaining a "royal flush"?
- b) what is the probability of obtaining a "pair"?

Answers

a)

There are $\binom{52}{5} = 2598960$ different 5-card hands.

$$P(\text{royal flush}) = \frac{4}{2598960} = 1.539077169 \times 10^{-6} \approx 0.0000015$$

(about one in a million chance)

b)

We denote the hand by AABCD, where AA corresponds to the equal-value pair of cards. Think for instance that A is ‘ace’. There are $\binom{4}{2} = 6$ ways of choosing. As there are 13 distinct values, the pair of cards can be chosen in $6 \times 13 = 78$ ways. The remaining three cards of the hand will be drawn out of the remaining 12 values, with each card being of any of the 4 suits. There are $\binom{12}{3}$ combinations of the values of the three cards. Since each of them can come from any of the 4 suits, we have

$$\binom{12}{3} \times 4 \times 4 \times 4 = 14080$$

different possibilities. Thus

$$P(\text{pair}) = 78 \times \frac{14080}{2598960} = 0.422569027 \approx 0.423$$

This is almost the probability of getting a ‘head’ in a fair coin flip (i.e. 0.5), or you can expect at least get one pair in a little more than 2 games.

The other probabilities of the Poker game can be found in https://en.wikipedia.org/wiki/Poker_probability

Example 4

Mark 6 consists of 49 numbers. 6 numbers are randomly drawn without replacement. Then an extra number is drawn without replacement.

4th price is won when you have guessed right on 4 numbers as well as the extra number.

Calculate the probability of winning the 4th price.

Notes about this question:

In Mark Six, people fill in 6 numbers only in the entry form. For the fourth price, 4 out of 6 numbers are correct, the 5th number is incorrect, and the 6th number must be the extra number.

https://en.wikipedia.org/wiki/Mark_Six

Prize	Criteria
1st	All 6 drawn numbers
2nd	5 out of 6 drawn numbers, plus the extra number
3rd	5 out of 6 drawn numbers
4th	4 out of 6 drawn numbers, plus the extra number
5th	4 out of 6 drawn numbers
6th	3 out of 6 drawn numbers, plus the extra number
7th	3 out of 6 drawn numbers

Answers

There are $\binom{49}{6}$ possible scenarios. There are $\binom{6}{4}$ ways of drawing the 4 correct numbers. There are $\binom{42}{1}$ ways to draw the incorrect number and there is one way to draw the extra number.

Hence the probability is

$$\frac{\binom{6}{4}\binom{42}{1}}{\binom{49}{6}} = 4.50520802 \times 10^{-5}$$

Example 5

Assume 45% of CityU undergraduates are male and 30% of CityU undergraduates are in the College of Engineering (CENG). Assume 70% of undergraduate students in CENG are male. If a student is selected at random, find the conditional probability that

- a) a CityU undergraduate student is male, given that the student is a CENG student.
- b) the student is in CENG, given that the student is male.

Answers

Let E be the event that an undergraduate student is male. Let F be the event that the student is in CENG.

$$P(E) = 0.45$$

$$P(F) = 0.3$$

a) $P(E|F) = 0.7$

b)

$$P(F|E) = \frac{P(EF)}{P(E)} = \frac{P(E|F)P(F)}{P(E)} = \frac{(0.7)(0.3)}{0.45} \approx 0.4667$$

Thus, a male undergraduate student only makes it slightly more likely for the student to be studying engineering.

Example 6

- a) Find the probability of a plane crash.
- b) If a plane crash occurs, find the probability of survival.
- c) Hence find the probability of dying in a plane crash.

Answers

Let F be the event that a plane crash occurs. Let E be the event that a passenger dies.

Find $P(E|F)$ and $P(F)$ from the web.

The probability of dying in a plane crash is

$$P(EF) = P(E|F)P(F)$$

Note: D. Carnegie, “How to Stop Worrying and Start Living” has a chapter titled “A law that will outlaw many of your worries”. He suggests not to over-worry about very unlikely events.

The book is available in CityU library.

Counter-intuitive Results (Apparent paradox) in Probability

Example 7

The Monte Hall Problem: <https://www.youtube.com/watch?v=mhlc7peGlGg>. Stop when the host gives you 10 seconds. Think about it carefully before continuing.

Suggest a way to make sense of the counter-intuitive result of the Monte Hall problem.

Answers

One possible way:

Let E_i be the event that the contestant opens door i

Let F_j be the event that the host opens door j , where $j \neq i$

E_i and F_j are not independent. F_j depends on E_i . So the event F_j reveals some subtle information.

There are many other ways to make sense of it. See

https://en.wikipedia.org/wiki/Monty_Hall_problem if interested.

Example 8

Assume that the probability of having a boy and a girl is equal and the events are independent.

- a) What is the probability that a family with two children has at least one boy?
- b) Suppose we know that the family has a boy, what is the probability that the other child is also a boy?
- c) Suppose we visit a family with two children. We knock the door. Assume that it is equally probable that the first or the second child show up. Suppose a boy shows up. What is the probability that the other child is also a boy?

Answers

a)

$$S = \{MF, FM, MM, FF\}$$

$$E = \{MF, FM, MM\}$$

$$P(E) = 3/4$$

b)

$$E_2 = \{MM\}$$

$$E_1 = \{MF, FM, MM\}$$

$$P(E_2|E_1) = 1/3$$

c)

Let * denotes the child that opens the door

$$S = \{M^*F, MF^*, F^*M, FM^*, M^*M, MM^*, F^*F, FF^*\}$$

$$E_2 = \{M^*M, MM^*\}$$

$$E_1 = \{M^*F, FM^*, M^*M, MM^*\}$$

$$P(E_2|E_1) = 1/2$$

Example 9

Assume for simplicity that a year has 365 days, i.e., we can ignore leap year. Suppose the probability that a person is born on any of the 365 days is equal and independent. There is a party of n persons. How large should n be such that the probability that two persons in the party have the same birthday is larger than 50%?

Answers

E : event that no person share the same birthday

$$P(E) = \left(\frac{365}{365}\right) \left(\frac{364}{365}\right) \cdots \left(\frac{365 - (n - 1)}{365}\right)$$

$$P(E) < 0.5 \text{ when } n \geq 23$$

Bayes' Rule

Example 10

In the example concerning laboratory blood test in the lecture notes, if the disease is very rare and 1 in 1 million (1×10^{-6}) chance of the population has the disease, what is the probability that a person has the disease given that the test result is positive?

Look up https://en.wikipedia.org/wiki/Rare_disease for the definition of rare disease.

Answers

$$P(E|F) = \frac{(0.95)(10^{-6})}{(0.95)(10^{-6}) + (0.01)(1 - 10^{-6})} \approx 95 \times 10^{-6}$$

The chance is increased 95 times but it is still very small.

Observe that the false positive probability is $10^{-2} \gg 10^{-6}$, the probability of having the disease. It suggests not to over-worry about very unlikely events.

Example 11

The following web page gives the statistics of cancer in Hong Kong:

<https://www3.ha.org.hk/cancereg/topten.html>

A patient comes to see a doctor and describes his symptoms. Based on the doctor's experience, he is 60% certain that the patient may have cancer. He asks the patient to conduct a test, which will give a cancer index number. A number that exceeds the threshold indicates cancer and vice versa.

Suppose that the test has a false positive rate of 10% and a false negative rate of 1%. What is the conditional probability that the patient has the cancer given that

- a) the test indicates that he has the cancer?
- b) the test indicates that he does not have the cancer?
- c) Do you think the test is useful? Comment.

Answers

Let E be the event that the patient has the cancer.

Let F be the event that the test indicates cancer, i.e., the test result is positive

Since the false negative rate is 1%,

$$P(F^c|E) = 0.01$$

Hence

$$P(F|E) = 0.99$$

Since the false positive rate is 10%

$$P(F|E^c) = 0.1$$

The doctor is 60% certain that the patient has the cancer. Thus

$$P(E) = 0.6$$

You may consider it as a subjective interpretation of probability. On the other hand, you may also interpret this objectively like this. The doctor has seen a large number of patients with the symptoms. The frequency for which patients having the symptoms have cancer is 0.6.

Note also that $P(F|E)$ describes the effectiveness of the test in identifying cancer when cancer is present. This probability is not related to the doctor's judgment.

$$P(E^c) = 1 - P(E) = 0.4$$

a)

$$P(E|F) = \frac{P(EF)}{P(F)} = \frac{P(F|E)P(E)}{P(F)} = \frac{(0.99)(0.6)}{P(F)}$$

$P(F)$ is the probability that the test indicates cancer. Using Bayes' rule,

$$P(F) = P(F|E)P(E) + P(F|E^c)P(E^c) = (0.99)(0.6) + (0.1)(0.4) = 0.634$$

Hence

$$P(E|F) = 0.936908517 = 93.69\%$$

b)

$$\begin{aligned} P(E|F^c) &= \frac{P(EF^c)}{P(F^c)} = \frac{P(F^c|E)P(E)}{1 - P(F)} = \frac{(1 - P(F|E))P(E)}{1 - P(F)} \\ &= \frac{(1 - 0.99)(0.6)}{(1 - 0.634)} = 0.016393442 \approx 1.64\% \end{aligned}$$

c)

The test is useful in this situation. It is good at confirming the doctor's judgement (probability increases from 60% to more than 93%). On the other hand, if the test result is negative, it is quite reassuring (probability decreases from 40% to less than 2%).

Example 12

At a certain stage of criminal investigation, the inspector in charge is 60% convinced of the guilt of a certain suspect. Suppose now that a new piece of evidence that shows the criminal has a certain characteristic is uncovered. If 20% of the population possesses this characteristic, how certain of the guilt of the suspect should the inspector now be if it turns out that the suspect has this characteristic?

Answers

Let E denote the event that the suspect is guilty and F the event that he possesses the characteristic of the criminal. We wish to calculate $P(E|F)$

$$P(E|F) = \frac{P(EF)}{P(F)}$$

$$P(EF) = P(F|E)P(E)$$

The inspector is 60% convinced that the suspect is the criminal. Hence $P(E) = 0.6$

As the new piece of evidence shows that the criminal has the characteristic,

$$P(F|E) = 1$$

$$\text{Hence } P(EF) = (1)(0.6) = 0.6$$

$$P(F) = P(F|E)P(E) + P(F|E^c)P(E^c)$$

As 20% of the population has this characteristic, $P(F|E^c) = 0.2$

$$P(F) = (1)(0.6) + (0.2)(0.4) = 0.68$$

Hence

$$P(E|F) = \frac{0.6}{0.68} = 0.882352941 \approx 0.882$$

Example 13

Let us now suppose that the new evidence is subject to different possible interpretations, and in fact only shows that it is 90% likely that the criminal possesses this certain characteristic. In this case, how likely would it be that the suspect is guilty (assuming, as before, that he has this characteristic)?

Answers

Now $P(F|E) = 0.9$

$$\begin{aligned} P(E|F) &= \frac{P(F|E)P(E)}{P(F|E)P(E) + P(F|E^c)P(E^c)} = \frac{(0.9)(0.6)}{(0.9)(0.6) + (0.2)(0.4)} \\ &= 0.870967741 \approx 0.8710 \end{aligned}$$

Example 14

EXAMPLE 3.7g A plane is missing and it is presumed that it was equally likely to have gone down in any of three possible regions. Let $1 - \alpha_i$ denote the probability the plane will be found upon a search of the i th region when the plane is, in fact, in that region, $i = 1, 2, 3$. (The constants α_i are called *overlook probabilities* because they represent the

3.8 Independent Events

75

probability of overlooking the plane; they are generally attributable to the geographical and environmental conditions of the regions.) What is the conditional probability that the plane is in the i th region, given that a search of region 1 is unsuccessful, $i = 1, 2, 3$?

SOLUTION Let R_i , $i = 1, 2, 3$, be the event that the plane is in region i ; and let E be the event that a search of region 1 is unsuccessful. From Bayes' formula, we obtain

$$\begin{aligned} P(R_1|E) &= \frac{P(ER_1)}{P(E)} \\ &= \frac{P(E|R_1)P(R_1)}{\sum_{i=1}^3 P(E|R_i)P(R_i)} \\ &= \frac{(\alpha_1)(1/3)}{(\alpha_1)(1/3) + (1)(1/3) + (1)(1/3)} \\ &= \frac{\alpha_1}{\alpha_1 + 2} \end{aligned}$$

For $j = 2, 3$,

$$\begin{aligned} P(R_j|E) &= \frac{P(E|R_j)P(R_j)}{P(E)} \\ &= \frac{(1)(1/3)}{(\alpha_1)1/3 + 1/3 + 1/3} \\ &= \frac{1}{\alpha_1 + 2}, \quad j = 2, 3 \end{aligned}$$

Thus, for instance, if $\alpha_1 = .4$, then the conditional probability that the plane is in region 1 given that a search of that region did not uncover it is $\frac{1}{6}$, whereas the conditional probabilities that it is in region 2 and that it is in region 3 are both equal to $\frac{1}{2.4} = \frac{5}{12}$. ■

Example 15

A binary classifier is one which classifies an input instance into “positive” or “negative”. The quality of a binary classifier can be expressed by its

- a) accuracy
- b) precision
- c) recall
- d) specificity

Find the definitions of these terms. What are the meanings of these terms?

Answers

#TP True Positive
#FP False Positive
#TN True Negative
#FN False Negative

$$Accuracy = \frac{\#TP + \#TN}{\#TP + \#FP + \#TN + \#FN}$$

It is the probability of making the correct decision.

$$Precision = \frac{\#TP}{\#TP + \#FP}$$

It is the probability that the diagnosis “positive” is accurate.

$$Recall = \frac{\#TP}{\#TP + \#FN}$$

Recall is also known as sensitivity, is the probability of detecting the positive.

$$Specificity = \frac{\#TN}{\#TN + \#FP}$$

Specificity is the probability of detecting the negative.

Detecting Independent Events

Example 16

Two fair die are thrown. Let E be the event that the second throw is 6 and let F be the event that the sum of the two throws is 10. Prove or disprove that the two events are independent.

Answers

$$P(E) = P\{(1,6), \dots (6,6)\} = \frac{6}{36} = \frac{1}{6}$$

$$P(F) = P\{(4,6), (5,5), (6,4)\} = \frac{3}{36} = \frac{1}{12}$$

$$P(EF) = P\{(4,6)\} = \frac{1}{36}$$

Since $P(E)P(F) \neq P(EF)$, the two events are not independent.

Example 17

Two fair dice are thrown. Let E be the event that the sum is 8. Let F be the event that the first throw is 2. Show that the two events are not independent.

Answers

$$P(E) = P\{(2,6), \dots, (6,2)\} = \frac{5}{36}$$

$$P(F) = P\{(2,1), \dots, (2,6)\} = \frac{1}{6}$$

$$P(EF) = P\{(2,6)\} = \frac{1}{36}$$

As $P(EF) \neq P(E)P(F)$, the two events are not independent.

Actually, as long as the sum is 8, the first nor the second dice cannot be 1.